

ROBUSTNESS OF THE DOUBLE DISCRIMINANT FUNCTION IN NONNORMAL SITUATIONS

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normal.

Summary : A study is made of the robustness of the Double
Discriminant Function introduced in Chang and Afifi
[*J. Amer. Statist. Assoc.* (1974), 336-339]. The
general results for the distribution functions of the
error rates as well as their expected values are derived.
Special cases of the truncated normal and mixtures of
normals (contamination model) are examined in
detail.

1. Introduction

Consider the vector $\mathbf{W}' = (X, Y)$ in which the random variable X has the
Bernoulli distribution

$$f(x, \theta) = \theta^x (1-\theta)^{1-x} \quad x = 0, 1$$

and Y , conditional on $X = x$, has the normal distribution with mean
 $\mu(x) = \mu + \delta x$ and variance $\sigma(x)^2 = \sigma^2 + \gamma^2 x$. The probability density
function of $\mathbf{W}$ can be written as

$$(1.1) \quad g(\mathbf{w}) = \theta^x (1-\theta)^{1-x} \left\{ \frac{1}{\sqrt{2\pi} \sigma(x)} \exp \left(-\frac{y-\mu(x)}{2\sigma(x)^2} \right) \right\}.$$

The expectation and variance matrix of the vector $\mathbf{W}$ are

$$E[\mathbf{W}] = \begin{bmatrix} \theta \\ \mu + \theta\delta \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \theta(1-\theta) & \theta(1-\theta)\delta \\ \theta^2 + \gamma^2\theta + \theta(1-\theta)\delta^2 & \end{bmatrix}.$$

The random variable Y has the probability density function

$$g^*(y) = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \frac{1}{\sqrt{2\pi} \sigma(x)} \exp \left\{ -\frac{1}{2\sigma(x)^2} (y-\mu(x))^2 \right\}$$

which is a mixture of two normal density functions.

Chang and Afifi (1974) examined the problem of classifying an observation $\mathbf{w}$ into one of two populations Π_1, Π_2 when Π_i has the pdf, for $i = 1, 2$,

$$g_i(\mathbf{w}) = \theta_i^x (1-\theta_i)^{1-x} \frac{1}{\sqrt{2\pi} \sigma_i(x)} \exp \left\{ -\frac{1}{2\sigma_i(x)^2} (y-\mu_i(x))^2 \right\}.$$

In what follows we assume that

$$\sigma_1(x)^2 = \sigma_2(x)^2, \text{ for } x = 0, 1.$$

Consider independent random samples of size n_i from Π_i , $i = 1, 2$. Let $y_{ij}^{(x)}$ be the j -th observation from the i -th population of which the X -value is x . Let

$$n_i = n_i^{(0)} + n_i^{(1)}$$

and

$$\bar{y}_i(x) = \frac{1}{n_i(x)} \sum_{j=1}^{n_i(x)} y_{ij}^{(x)},$$

$$s(x)^2 = \frac{1}{n_1^{(x)} + n_2^{(x)}} \sum_{i=1}^2 n_i^{(x)} \sum_{j=1}^{n_i^{(x)}} [y_{ij}^{(x)} - \bar{y}_i^{(x)}]^2,$$

It can be shown that the maximum likelihood classification rule is to classify $\mathbf{w}$ as belonging to Π_1 or Π_2 according as

$$(1.2) \quad Z(x) = \left[y - \frac{1}{2}(\bar{y}_1^{(x)} + \bar{y}_2^{(x)}) \right] \frac{(\bar{y}_1^{(x)} - \bar{y}_2^{(x)})}{s(x)^2} + \ln \left\{ \frac{n_1^{(x)} n_2}{n_2^{(x)} n_1} \right\} \gtrless 0.$$

From (1.2) we see that we have two discriminant functions depending upon the observed value of X . This is referred to as the Double Discriminant Function.

Balakrishnan *et al.* (1986a) have studied the behaviour of the errors of misclassification when $\theta_1 = \theta_2$; they have derived the distribution functions as well as the expected values of these errors. The present paper is concerned with the robustness of the double discriminant procedure when the random variable Y has a nonnormal distribution. Using the techniques developed by Kocherlakota *et al.* (1986b), general formulas are obtained. These are exemplified by particular situations: Truncated normal and the mixtures of normals (contaminated normal distribution). Detailed results are given in these two cases.

2. Errors of misclassification

The classification rule specified by (1.2) becomes, under the assumption $\theta_1 = \theta_2$,

classify $\mathbf{w}$ to Π_1 if

$$Z(x) = \left[y - \frac{1}{2}(\bar{y}_1(x) + \bar{y}_2(x)) \right] (\bar{y}_1(x) - \bar{y}_2(x)) > 0$$

and to Π_2 otherwise. Thus the errors of misclassification for $X = x$ are

$$e_{12}^{(x)} = P\{Z(x) < 0 | w \in \Pi_1\}$$

and

$$e_{21}^{(x)} = P\{Z(x) > 0 | w \in \Pi_2\}.$$

Or, substituting for $Z(x)$, we can write $e_{12}^{(x)}$ as

$$\begin{aligned} (2.1) \quad e_{12}^{(x)} &= P\left\{Y - \frac{1}{2}(\bar{Y}_1(x) + \bar{Y}_2(x)) < 0 | w \in \Pi_1\right\} \text{ if } \bar{Y}_1(x) > \bar{Y}_2(x) \\ &= P\left\{Y - \frac{1}{2}(\bar{Y}_1(x) + \bar{Y}_2(x)) > 0 | w \in \Pi_1\right\} \text{ if } \bar{Y}_1(x) < \bar{Y}_2(x) \end{aligned}$$

with a similar expression for $e_{21}^{(x)}$.

2.1 The distribution function

From this it is obvious that e_{12} (as well as e_{21}) is a random variable with its distribution depending upon the population from which we are sampling. Let, for $X = x$, the distribution function of Y be $F_1^{(x)}(t)$ and the pdf be $f_1^{(x)}(t)$, for the population Π_1 . Then the distribution function of e_{12} is

$$(2.2) \quad G_1(z) = P\{e_{12} \leq z\}$$

$$= \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} P\{e_{12}^{(x)} \leq z\}$$

$$\begin{aligned}
&= \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[P\left\{F_1^{(x)}\left(\frac{U^{(x)}}{2}\right) \leq z, V^{(x)} > 0\right\} \right. \\
&\quad \left. + P\left\{1 - F_1^{(x)}\left(\frac{U^{(x)}}{2}\right) \leq z, V^{(x)} \leq 0\right\} \right]
\end{aligned}$$

where

$$U^{(x)} = Y_1^{(x)} + Y_2^{(x)}, \quad V^{(x)} = Y_1^{(x)} - Y_2^{(x)}.$$

As has been shown in Kocherlakota *et al.* (1986b) it is possible to consider two cases in evaluating $G_I(z)$, explicitly :

(i) If $F_1^{(x)-1}(p)$, $0 < p < 1$, is given explicitly, then

$$\begin{aligned}
(2.3) \quad G_I(z) &= \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[P\left\{U^{(x)} \leq 2F_1^{(x)-1}(z), V^{(x)} > 0\right\} \right. \\
&\quad \left. + P\left\{U^{(x)} \geq 2F_1^{(x)-1}(1-z), V^{(x)} \leq 0\right\} \right]
\end{aligned}$$

From the joint distribution of $(U^{(x)}, V^{(x)})$, we can determine $G_I(z)$.

(ii) If $F_1^{(x)-1}(p)$ cannot be determined explicitly, we can expand

$F_1\left(\frac{U^{(x)}}{2}\right)$ in a series involving $U^{(x)}$. Then the distribution function of e_{12} is given approximately by

$$(2.4) \quad G_1(z) = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[P \left\{ U(x) \leq \frac{z - (1-\kappa_1^{(x)})}{\kappa_2^{(x)}}, \quad V(x) \geq 0 \right\} \right. \\ \left. + P \left\{ U(x) > \frac{\kappa_1^{(x)} - z}{\kappa_2^{(x)}}, \quad V(x) \leq 0 \right\} \right]$$

where

$$\kappa_1^{(x)} = 1 - F_1^{(x)}(0), \quad \kappa_2^{(x)} = f_1^{(x)}(0)/2.$$

It is possible to express $G_2(z)$, the distribution function of e_{21} , in a similar manner.

2.2 Expected value

Since $e_{12}^{(x)}$ is given, for $X=x$, by (2.1), we can write the unconditional probability of misclassification (or the expected value of e_{12}) as

$$(2.5) \quad E[e_{12}] = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} E[e_{12}^{(x)}] \\ = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[P \left\{ Y < \frac{U(x)}{2} \mid \bar{Y}_1^{(x)}, \bar{Y}_2^{(x)} \right\} P \left\{ \bar{Y}_1^{(x)} > \bar{Y}_2^{(x)} \right\} \right. \\ \left. + P \left\{ Y > \frac{U(x)}{2} \mid \bar{Y}_1^{(x)}, \bar{Y}_2^{(x)} \right\} P \left\{ \bar{Y}_1^{(x)} \leq \bar{Y}_2^{(x)} \right\} \right]$$

or

$$(2.6) \quad E[e_{12}] = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[\int P \left\{ U(x) > 2y, \quad V(x) > 0 \right\} f_1^{(x)}(y) dy \right] \\ + \int P \left\{ U(x) \leq 2y, \quad V(x) \leq 0 \right\} f_1^{(x)}(y) dy \Big].$$

The probabilities in the integrals on the right-hand side of (2.6) are evaluated from the joint distribution of $(U^{(x)}, V^{(x)})$.

A similar expression for $E[e_{2I}]$ is obtained in terms of the joint distribution of $(U^{(x)}, V^{(x)})$.

3. Truncated normal

The behaviour of the linear discriminant function when used for the truncated normal distribution has been studied by Kocherlakota *et al.* (1986a). Such a model for nonnormality seems appropriate as the distribution resembles the normal distribution while allowing us to induce nonnormality by the excision of the tails.

Here we take the distribution of Y , given $X=x$, to be given by

$$(3.1) \quad g^{(x)}(y) = \frac{1}{\sqrt{2\pi}} \frac{e^{-\frac{1}{2}z^2}}{[\Phi(b) - \Phi(a)]}, \quad a < z < b,$$

where

$$z = (y - \mu^{(x)})/\sigma^{(x)}, \quad \mu^{(x)} = \mu + \delta x, \quad \sigma^{(x)^2} = \sigma^2 + \gamma^2 x.$$

It should be noted that in (3.1) the distribution of Y has truncation points which are functions of x .

For this distribution, since the standardized variable has the pdf. (3.1), we see that $F^{(x)}(t_I) = z$ can be inverted from

$$\frac{\Phi\left[\frac{t_I^{(x)} - \mu^{(x)}}{\sigma^{(x)}}\right] - \Phi(a)}{\Phi(b) - \Phi(a)} = z$$

to yield

$$(3.2a) \quad t_I^{(x)} = \mu^{(x)} + \sigma^{(x)} \Phi^{-1}[\Phi(a) + z\{\Phi(b) - \Phi(a)\}].$$

Similarly, $F^{(x)}(t_2) = 1-z$ has the inverse

$$(3.2b) \quad t_2^{(x)} = \mu^{(x)} + \sigma^{(x)} \Phi^{-1}[\Phi(b) - z\{\Phi(b) - \Phi(a)\}].$$

The sample mean $\bar{Y}_j^{(x)}$, based on a random sample of $n_j^{(x)}$ from the population Π_j , $j = 1, 2$, does not have the normal distribution. However, using the Central Limit Theorem it is possible to write

$$\bar{Y}_j^{(x)} \approx N\left[\tau_j^{(x)}, \eta^{(x)2}/n_j^{(x)}\right]$$

where

$$\tau_j^{(x)} = \mu_j^{(x)} + \sigma^{(x)} \xi,$$

$$\eta^{(x)2} = \sigma^{(x)2} \left[1 - \left\{ \frac{b\Phi(b) - a\Phi(a)}{\Phi(b) - \Phi(a)} \right\} - \xi^2 \right]$$

with

$$\xi = \{\Phi(a) - \Phi(b)\} / \{\Phi(b) - \Phi(a)\}.$$

Hence $U^{(x)}$ and $V^{(x)}$ can be considered to have asymptotically the bivariate normal distribution with:

$$E[U^{(x)}] = \tau_1^{(x)} + \tau_2^{(x)} = \tau^{(x)},$$

$$E[V^{(x)}] = \tau_1^{(x)} - \tau_2^{(x)} = \delta^{(x)} = \mu_1^{(x)} - \mu_2^{(x)},$$

$$\sigma_{U^{(x)}}^2 = \sigma_{V^{(x)}}^2 = \alpha^{(x)2} = \eta^{(x)2} \left[\frac{1}{n_1^{(x)}} + \frac{1}{n_2^{(x)}} \right],$$

$$\sigma_{U^{(x)}, V^{(x)}} = \eta^{(x)2} \left[\frac{1}{n_1^{(x)}} - \frac{1}{n_2^{(x)}} \right],$$

and

$$\rho_{U(x), V(x)} = \left[\frac{n_2^{(x)} - n_1^{(x)}}{n_1^{(x)} n_2^{(x)}} \right] = \rho^{(x)}.$$

3.1 Distribution function

Assuming the asymptotic distribution of $(U^{(x)}, V^{(x)})$ to be bivariate normal, it is possible to write the distribution function of e_{12} , using equation (2.3). Let $H(z_1, z_2; \rho)$ be the distribution function of the standard bivariate normal with the coefficient of correlation ρ .

Then

$$(3.3) \quad G_1(z) = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[H \left\{ \frac{2t_1^{(x)} - \tau^{(x)}}{\alpha^{(x)}}, \frac{\delta^{(x)}}{\alpha^{(x)}}; -\rho^{(x)} \right\} \right. \\ \left. + H \left\{ \frac{\tau^{(x)} - 2t_2^{(x)}}{\alpha^{(x)}}, -\frac{\delta^{(x)}}{\alpha^{(x)}}; -\rho^{(x)} \right\} \right],$$

where $t_1^{(x)}$ and $t_2^{(x)}$ are given by (3.2a) and (3.2b) respectively, replacing $\mu^{(x)}$ by $\mu_1^{(x)}$.

A similar expression can be given for the distribution function $G_2(z)$ of e_{21} .

$$(3.4) \quad G_2(z) = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[H \left\{ \frac{\tau^{(x)} - 2t_1^{*(x)}}{\alpha^{(x)}}, -\frac{\delta^{(x)}}{\alpha^{(x)}}; \rho^{(x)} \right\} \right. \\ \left. + H \left\{ \frac{2t_2^{*(x)} - \tau^{(x)}}{\alpha^{(x)}}, -\frac{\delta^{(x)}}{\alpha^{(x)}}; \rho^{(x)} \right\} \right]$$

where $t_1^{*(x)}$ and $t_2^{*(x)}$ are obtained from (3.2a) and (3.2b) respectively, replacing $\mu^{(x)}$ by $\mu_2^{(x)}$.

3.2 Expected value

Using the distribution of $(U^{(x)}, V^{(x)})$ as above, we can write the integrals involved in (2.6) as

$$(3.5) \quad \int_{\mu_1^{(x)} + a\sigma(x)}^{\mu_1^{(x)} + b\sigma(x)} H\left[\frac{\tau(x) - 2y}{\alpha(x)}, \frac{\delta(x)}{\alpha(x)}; \rho(x)\right] f_1^{(x)}(y) dy \\ + \int_{\mu_1^{(x)} + a\sigma(x)}^{\mu_1^{(x)} + b\sigma(x)} H\left[\frac{2y - \tau(x)}{\alpha(x)}, -\frac{\delta(x)}{\alpha(x)}; \rho(x)\right] f_1^{(x)}(y) dy.$$

Performing the integrations by parts we use the result

$$\int_{-\infty}^y \Phi[\alpha + \beta x] \Phi[\gamma + \delta x] \Phi(x) dx \\ = T\left[\frac{\gamma}{\sqrt{1 + \delta^2}}, \frac{\alpha}{\sqrt{1 + \beta^2}}, y; \frac{\beta\delta}{\sqrt{(1 + \beta^2)(1 + \delta^2)}}, -\frac{\beta}{\sqrt{1 + \beta^2}}, -\frac{\delta}{\sqrt{1 + \delta^2}}\right],$$

where $T[z_1, z_2, z_3; \rho_{12}, \rho_{13}, \rho_{23}]$ stands for the distribution function of a standard trivariate normal distribution.

Thus

$$(3.6) \quad E[e_{12}] = \frac{1}{\Phi[b] - \Phi[a]} \sum_{x=0}^1 [k_1^{(x)}(b) - k_1^{(x)}(a)] \theta^x (1 - \theta)^{1-x}$$

where

$$(3.7) \quad k_1^{(x)}(t) = \Phi(t) \{ H[A^{(x)}, -B^{(x)}; \rho^{(x)}] + H[-A^{(x)}, B^{(x)}; \rho^{(x)}] \} \\ - T[-A^{(x)}, C^{(x)}, B^{(x)}; -\rho^{(x)} \rho^{*(x)}, -\rho^{*(x)}, \rho^{(x)}] \\ + T[A^{(x)}, C^{(x)}, B^{(x)}; \rho^{(x)} \rho^{*(x)}, -\rho^{*(x)}, -\rho^{(x)}]$$

where

$$A(x) = \frac{\delta(x)}{\alpha(x)}, \quad B(x) = \frac{2\sigma(x)t + 2\mu_1(x) - \tau(x)}{\alpha(x)},$$

$$C(x) = [\tau(x) - 2\mu_1(x)]/[\alpha(x)^2 + 4\sigma(x)^2]^{\frac{1}{2}},$$

$$\rho^*(x) = \alpha(x)/[\alpha(x)^2 + 4\sigma(x)^2]^{\frac{1}{2}}.$$

Similarly

$$(3.8) \quad E[e_{2I}] = \frac{1}{\Phi(b) - \Phi(a)} \sum_{x=a}^b [k_2^{(x)}(b) - k_2^{(x)}(a)] \theta^x (1-\theta)^{1-x}$$

where

$$(3.9) \quad k_2^{(x)}(t) = \Phi(t) [H\{A^{(x)}, B_1^{(x)}, -\rho^{(x)}\} + H\{-A^{(x)}, -B_1^{(x)}, -\rho^{(x)}\}]$$

$$- T[A^{(x)}, C_1^{(x)}, B_1^{(x)}, \rho^*(x)\rho(x), -\rho^*(x), -\rho(x)]$$

$$+ T[-A^{(x)}, C_1^{(x)}, B_1^{(x)}, -\rho^*(x)\rho(x), -\rho^*(x), \rho(x)]$$

with

$$B_1^{(x)} = \frac{2\sigma(x)t + 2\mu_2(x) - \tau(x)}{\alpha(x)},$$

$$C_1^{(x)} = [\tau(x) - 2\mu_2(x)]/(\alpha(x)^2 + 4\sigma(x)^2)^{\frac{1}{2}}.$$

3.3 Numerical results

In this section we have examined the effects of the conditional distribution of Y given $X = x$ being a truncated normal under Π_1 and Π_2 .

Although Z , the standardized variable in each case has truncation points independent of x , the actual distribution of Y itself has the truncation points as functions of x . A small selection of truncations as well as the

parameter space is presented here ; a wider set is presented in Balakrishnan *et al.* (1986b). The effects of nonnormality are examined through the distribution function $G_1(z)$, $G_2(x)$ and the expected values of the error rates.

It would be appropriate to indicate the models and the parameter space considered in the normal situation by Balakrishnan *et al.* (1986a). The parameters of the distribution of Y given $X = x$, for population j are $\mu_j^{(x)} = \mu_j + x\delta$, $\sigma_j^{(x)^2} = \sigma_j^2 + \gamma^2 x$; that is :

	<u>Population I</u>	<u>Population II</u>
$x = 0$	$\mu_1; \sigma^2$	$\mu_2; \sigma^2$
$x = 1$	$\mu_1 + \delta; \sigma^2 + \gamma^2$	$\mu_2 + \delta; \sigma^2 + \gamma^2$

We note that the difference $\delta^{(x)}$ between the population means remains $(\mu_1 - \mu_2)$ under $x = 0$ and $x = 1$. Also the variances are equal in the two populations under $x = 0$ and $x = 1$. Four parametric combinations are considered there:

Model I	$\mu_1 = 0$,	$\mu_2 = 1$,	$\delta = 0,5$,	$\sigma^2 = 1$,	$\gamma^2 = 1$,
Model II	$\mu_1 = 0$,	$\mu_2 = 2$,	$\delta = 0,5$,	$\sigma^2 = 1$,	$\gamma^2 = 1$,
Model III	$\mu_1 = 0$,	$\mu_2 = 1$,	$\delta = 0,5$,	$\sigma^2 = 4$,	$\gamma^2 = 1$,
Model IV	$\mu_1 = 0$,	$\mu_2 = 1$,	$\delta = 0,5$,	$\sigma^2 = 1$,	$\gamma^2 = 4$.

Model I is essentially the standard model with the mean of Π_2 shifted to the right. Model II has this shift increased. Models III and IV are similar to Model I with the exception that the variances are changed. It should be noted that the results are given for $n_j^{(0)} = n_j^{(1)}$, $j = 1, 2$. Here these models are considered when the conditional distribution of Y is

truncated at $(-1,65, 1,65)$ and $(-1,28, 1,65)$.

The computations of the bivariate normal distribution function are done using the IMSL subroutine MDBNOR ; and those of the trivariate normal distribution function are done using the subroutine MULNOR of Schervish (1984).

- (i) Distribution function. Figures 1–4 depict the behaviour of $G_1(z)$ and $G_2(z)$ for the various cases described above and with $\theta = 0,5$. More extensive results are available in Balakrishnan, *et al.* (1986b, 1986c). It should be noted that, under symmetric truncation with the assumption of equality of variances, e_{12} and e_{21} are identically distributed. As can be seen from the graphs there is little difference in the pattern of behaviour of the two types of errors. An important factor that emerges is that as the truncation is increased the error rates decrease. This is evidenced by the sharp rise in the distribution function closer to the origin. This is no doubt caused by an increased separation of the populations due to truncation.
- (ii) Expectation. Table I gives the expected values of the error rates under various models and truncations. Our results indicate that as in the case of the distribution function an increase in the truncation reduces both $E[e_{12}]$ and $E[e_{21}]$. An increase in θ increases both of the expectations. However, an increase in n has little or no effect on the expected values. One point that should be noted is that when the truncation is symmetric $E[e_{12}] = E[e_{21}]$. For the asymmetric truncation $E[e_{21}] > E[e_{12}]$.

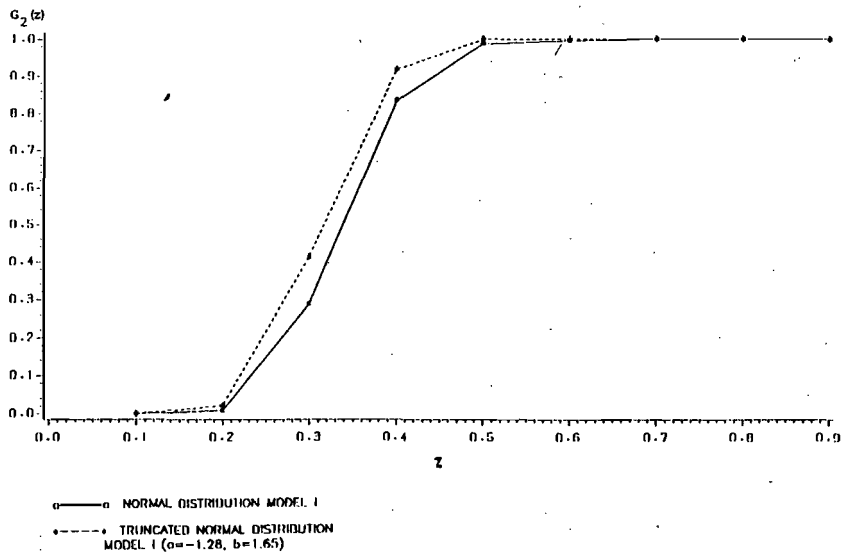
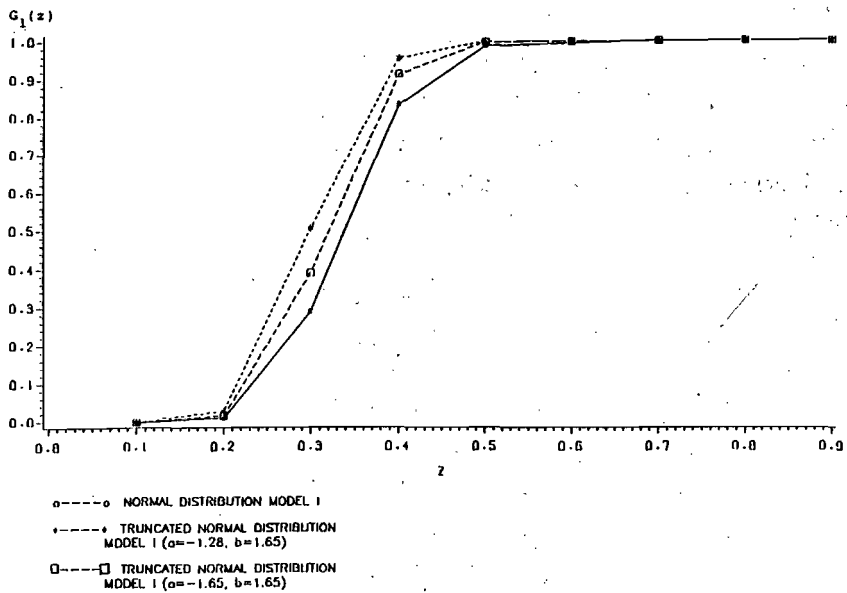


Figure 1. Distribution function $G_1(z)$ and $G_2(z)$ for $\theta = 0.50$ and $n_1 = n_2 = 40$

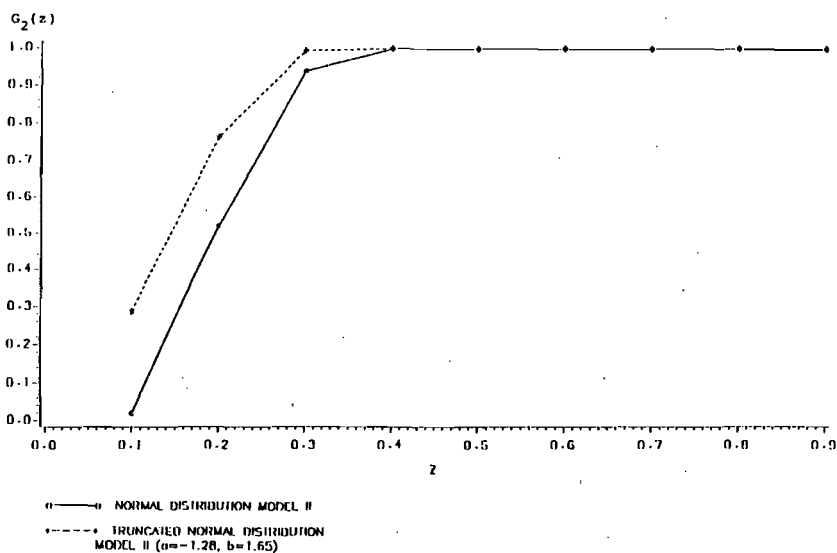
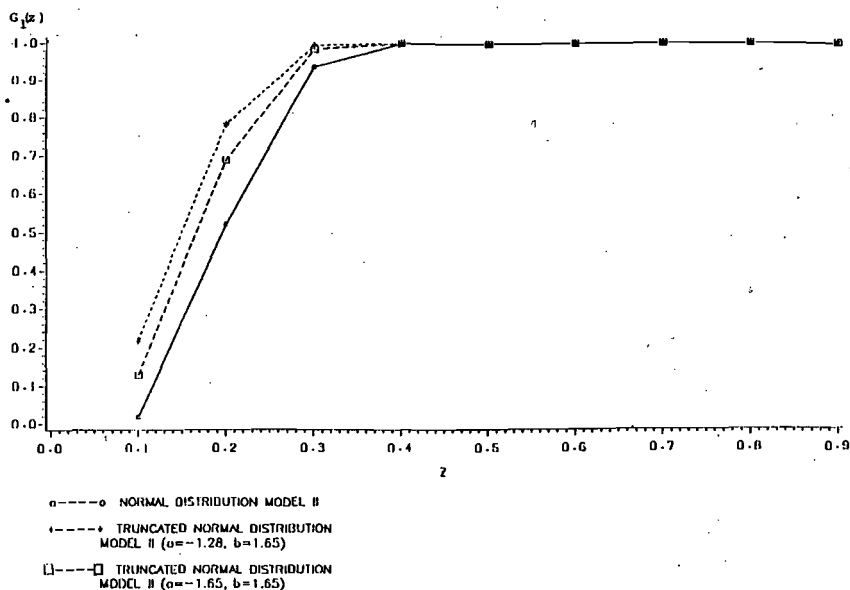
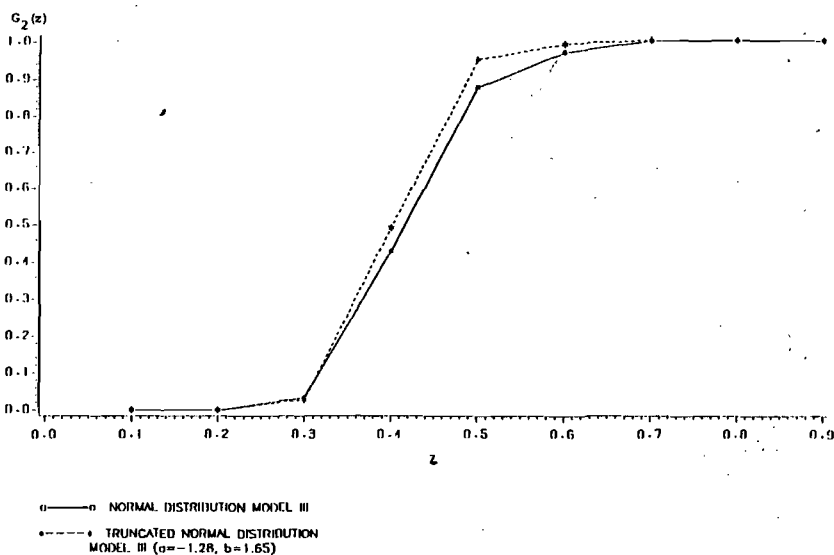
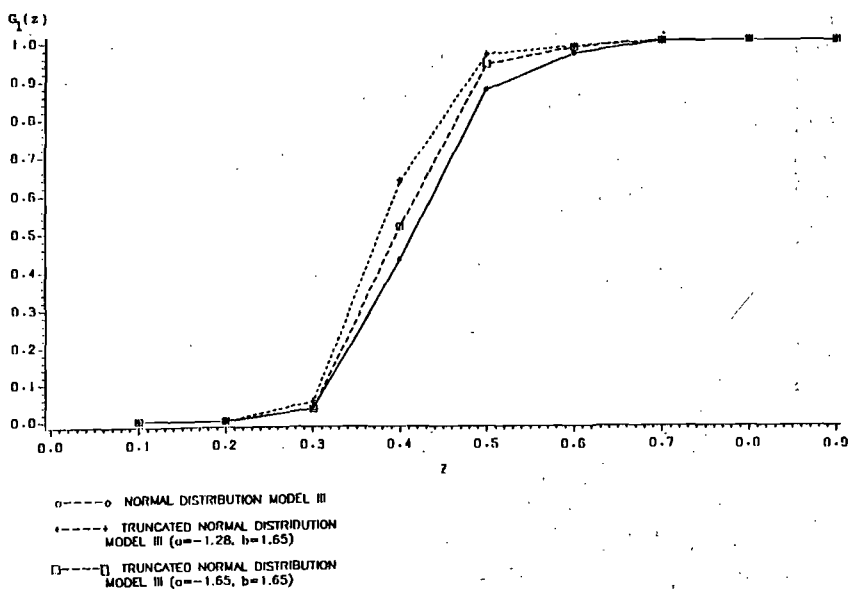


Figure 2. Distribution function $G_1(z)$ and $G_2(z)$ for $\theta = 0,50$ and $n_1 = n_2 = 40$

Figure 3. Distribution function $G_1(z)$ and $G_2(z)$ for $\theta = 0.50$ and $n_1 = n_2 = 40$

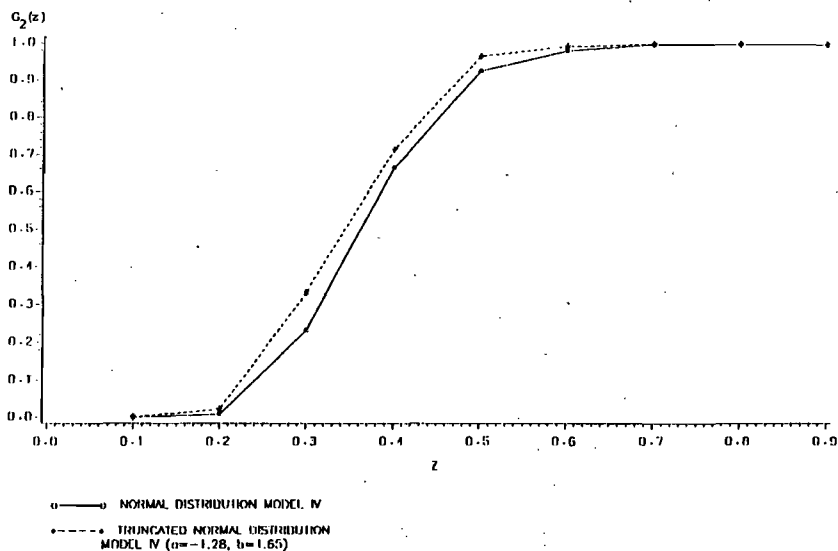
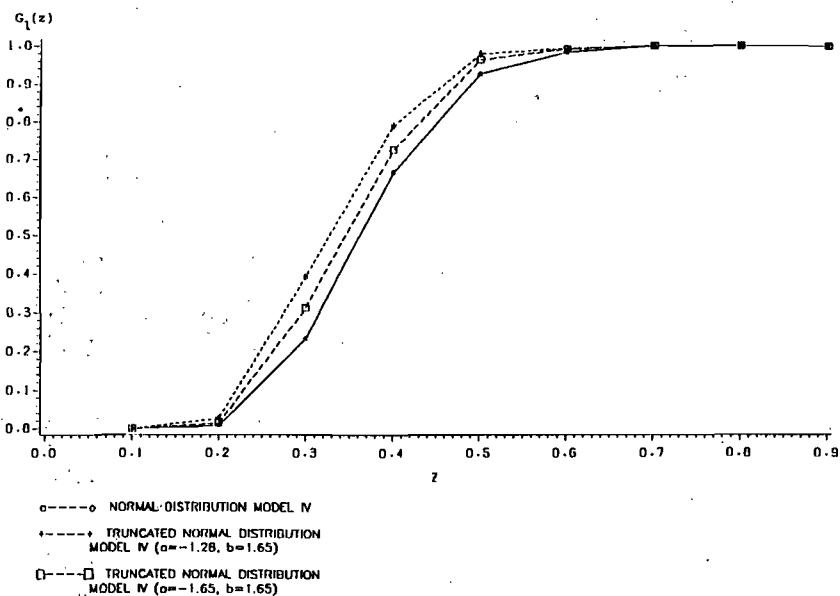


Figure 4. Distribution function $G_1(z)$ and $G_2(z)$ for $\theta = 0,50$ and $n_1 = n_2 = 40$

TABLE 1. Expected values of e_{12} and e_{21}
(Truncated normal distribution)
 $n_1 = n_2 = 20$

θ/Model	Normal*			
	I	II	III	IV
0,25	0,333	0,185	0,432	0,348
0,50	0,349	0,205	0,435	0,379
0,75	0,365	0,225	0,438	0,410
$a = -1,65, b = 1,65^*$				
0,25	0,308	0,148	0,413	0,324
0,50	0,324	0,170	0,416	0,357
0,75	0,340	0,192	0,420	0,390
$a = -1,28, b = 1,65$				
0,25	0,289	0,130	0,396	0,307
	0,301	0,122	0,412	0,319
0,50	0,305	0,153	0,400	0,340
	0,318	0,148	0,415	0,353
0,75	0,321	0,175	0,403	0,374
	0,335	0,174	0,419	0,388
$n_1 = n_2 = 40$				
Normal*				
0,25	0,325	0,182	0,417	0,340
0,50	0,339	0,202	0,420	0,369
0,75	0,353	0,222	0,423	0,398
$a = -1,65, b = 1,65^*$				
0,25	0,304	0,146	0,400	0,319
0,50	0,319	0,168	0,403	0,349
0,75	0,334	0,191	0,406	0,380
$a = -1,28, b = 1,65$				
0,25	0,286	0,129	0,383	0,302
	0,299	0,122	0,401	0,315
0,50	0,301	0,151	0,386	0,332
	0,315	0,147	0,404	0,347
0,75	0,316	0,173	0,390	0,363
0,75	0,331	0,173	0,408	0,379

$n_1 = n_2 = 100$				
Normal*				
0,25	0,323	0,180	0,406	0,336
0,50	0,336	0,200	0,408	0,362
0,75	0,349	0,221	0,411	0,388
$a = -1,65, b = 1,65^*$				
0,25	0,303	0,144	0,394	0,317
0,50	0,318	0,167	0,397	0,345
0,75	0,332	0,189	0,400	0,374
$a = -1,28, b = 1,65$				
0,25	0,285	0,128	0,378	0,300
	0,298	0,121	0,397	0,313
0,50	0,300	0,150	0,381	0,329
	0,314	0,146	0,400	0,344
0,75	0,315	0,172	0,384	0,358
	0,330	0,172	0,403	0,375

* Indicates that $E[e_{12}] = E[e_{21}]$. In other cases upper entry is $E[e_{12}]$ and lower is $E[e_{21}]$.

4. Mixtures of normals

Let the conditional distribution of Y given $X = x$ for the population Π_i be for $i = 1, 2$,

$$(4.1) \quad g_i^{(x)}(y) = p\phi(y, \mu_i^{(x)}, \sigma^{(x)^2}) + (1-p)\phi(y, \mu_i^{(x)} + a\sigma^{(x)}, \sigma^{(x)^2})$$

where

$$\mu_i^{(x)} = \mu_i + \delta x, \quad \sigma^{(x)^2} = \sigma^2 + \gamma^2 x.$$

This model is appropriate in situations where it is suspected that a population is contaminated by the presence of observations from another. Such a consideration of nonnormality was first discussed by Tukey (1960). The practical implications of the model are obvious, particularly in the problems involving classification.

The problem being studied is the robustness of the double discriminant function (1.2) when applied to the situations defined by (4.1). It can be seen that the errors of misclassification become:

$$(4.2) \quad e_{12}^{(x)} = p\Phi\left[\frac{2\mu_1^{(x)} - U^{(x)}}{2\sigma^{(x)}}\right] + (1-p)\Phi\left[\frac{2a\sigma^{(x)} + 2\mu_1^{(x)} - U^{(x)}}{2\sigma^{(x)}}\right] \quad \text{if } V^{(x)} < 0$$

$$= p\Phi\left[\frac{U^{(x)} - 2\mu_1^{(x)}}{2\sigma^{(x)}}\right] + (1-p)\Phi\left[\frac{U^{(x)} - 2\mu_1^{(x)} - 2a\sigma^{(x)}}{2\sigma^{(x)}}\right] \quad \text{if } V^{(x)} \geq 0$$

and a similar expression for $e_{21}^{(x)}$.

4.1 Distribution function

From equation (2.4) in section 2, the distribution function of e_{12} is given approximately by

$$(4.3) \quad G_1(z) = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \left[P\left\{ U^{(x)} \leq \frac{z - (1 - \kappa_1^{(x)})}{\kappa_2^{(x)}}, \quad V^{(x)} \geq 0 \right\} \right. \\ \left. + P\left\{ U^{(x)} \geq \frac{\kappa_1^{(x)} - z}{\kappa_2^{(x)}}, \quad V^{(x)} < 0 \right\} \right]$$

with

$$\kappa_1^{(x)} = 1 - \left[p\Phi\left(\frac{\mu_1^{(x)}}{\sigma^{(x)}}\right) + (1-p)\Phi\left(\frac{\mu_1^{(x)}}{\sigma^{(x)}} + a\right) \right]$$

$$\kappa_2^{(x)} = \frac{1}{2\sigma^{(x)}} \left[p\Phi\left(\frac{\mu_1^{(x)}}{\sigma^{(x)}}\right) + (1-p)\Phi\left(\frac{\mu_1^{(x)}}{\sigma^{(x)}} + a\right) \right]$$

The joint distribution of $U^{(x)}$ and $V^{(x)}$ can be derived from the results of Subrahmaniam (1972). It is seen to be bivariate normal with the parameters, for fixed r and s , where $r = 0, 1, \dots, n_1(x)$ and $s = 0, 1, \dots, n_2(x)$.

$$E[U^{(x)}] = \lambda_{1r}^{(x)} + \lambda_{2s}^{(x)}, \quad E[V^{(x)}] = \lambda_{1r}^{(x)} - \lambda_{2s}^{(x)},$$

$$\text{Var}[U^{(x)}] = \text{Var}[V^{(x)}] = \sigma^{(x)^2} \left(\frac{1}{n_1^{(x)}} + \frac{1}{n_2^{(x)}} \right),$$

$$\text{Cov}(U^{(x)}, V^{(x)}) = \sigma^{(x)^2} \left[\frac{n_2^{(x)} - n_1^{(x)}}{n_1^{(x)} + n_2^{(x)}} \right],$$

$$\rho_{U^{(x)}, V^{(x)}} = \frac{n_2^{(x)} - n_1^{(x)}}{n_1^{(x)} + n_2^{(x)}} = \rho^{(x)},$$

$$\lambda_{1r}^{(x)} = \mu_1^{(x)} + \left\{ 1 - \frac{r}{n_1^{(x)}} \right\} a \sigma^{(x)}$$

and

$$\lambda_{2r}^{(x)} = \mu_2^{(x)} + \left\{ 1 - \frac{s}{n_2^{(x)}} \right\} a \sigma^{(x)}.$$

Substituting for this joint distribution and representing the distribution function of the SBVN(ρ) by $H(z_1, z_2; \rho)$, the distribution function $G_1(z)$ is:

$$\begin{aligned} (4.4) \quad G_1(z) = & \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \sum_{r=0}^{n_1^{(x)}} \sum_{s=0}^{n_2^{(x)}} \binom{n_1^{(x)}}{r} \binom{n_2^{(x)}}{s} \\ & \cdot p^{r+s} (1-p)^{n_1^{(x)}+n_2^{(x)}-r-s} \cdot \{ H \alpha_1^{(x)}, \beta_1^{(x)}; -\rho^{(x)} \} \\ & + H(\alpha_2^{(x)}, -\beta_1^{(x)}; -\rho^{(x)}) \} \end{aligned}$$

where

$$\alpha_1^{(x)} = \left\{ \frac{z - (1 - \kappa_1^{(x)})}{\kappa_2^{(x)}} - \lambda_{1r}^{(x)} - \lambda_{2s}^{(x)} \right\} / \sigma^{*(x)},$$

$$\beta_1^{(x)} = \left\{ \lambda_{1r}^{(x)} - \lambda_{2s}^{(x)} \right\} / \sigma^{*(x)},$$

$$\alpha_2^{(x)} = \left\{ \lambda_{1r}^{(x)} + \lambda_{2s}^{(x)} - \frac{\kappa_1^{(x)} - z}{\kappa_2^{(x)}} \right\} / \sigma^{*(x)},$$

with

$$\sigma^{*(x)2} = \sigma^{(x)2} \left(\frac{1}{n_1^{(x)}} + \frac{1}{n_2^{(x)}} \right).$$

An expression for $G_2(z)$ can be obtained in a similar fashion.

4.2 Expected value

From (2.5) we can write

$$(4.5) \quad \begin{aligned} E[e_{12}] &= \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \{ P[U^{(x)} < 2Y, V^{(x)} < 0] \\ &\quad + P[U^{(x)} > 2Y, V^{(x)} > 0] \}. \end{aligned}$$

Substituting for the joint distribution of $(U^{(x)}, V^{(x)})$, for fixed r, s , and y , it can be seen that

$$P[U^{(x)} < 2y, V^{(x)} < 0] = H \left[\frac{2y - (\lambda_{1r}^{(x)} + \lambda_{2s}^{(x)})}{\sigma^{*(x)}}, \frac{\lambda_{2s}^{(x)} - \lambda_{1r}^{(x)}}{\sigma^{*(x)}}; \rho^{(x)} \right]$$

and

$$P[U^{(x)} > 2y, V^{(x)} > 0] = H \left[\frac{\lambda_{1r}^{(x)} + \lambda_{2s}^{(x)} - 2y}{\sigma^{*(x)}}, \frac{\lambda_{1r}^{(x)} - \lambda_{2s}^{(x)}}{\sigma^{*(x)}}; \rho^{(x)} \right].$$

Integrating with respect to the pdf of y , for fixed r, s and x :

$$P[U^{(x)} < 2y, V^{(x)} < 0] = pH[\eta_1^{(x)}, \beta_1^{(x)}; \eta_2^{(x)}] + (1-p)H[\eta_3^{(x)}, \beta_1^{(x)}; \eta_2^{(x)}]$$

and

$$P[U^{(x)} > 2y, V^{(x)} > 0] = pH[\eta_1^{(x)}, -\beta_1^{(x)}; -\eta_2^{(x)}] + (1-p)H[\eta_3^{(x)}, -\beta_1^{(x)}; -\eta_2^{(x)}],$$

where the new constants appearing here are:

$$\eta_1^{(x)} = \left[4 + \frac{1}{n_1^{(x)}} \frac{1}{n_2^{(x)}} \right]^{-\frac{1}{2}} \left[\frac{2\mu_1^{(x)} - (\lambda_{1r}^{(x)} + \lambda_{2s}^{(x)})}{\sigma^{(x)}} \right],$$

$$\eta_2^{(x)} = \left\{ \frac{n_1^{(x)} + n_2^{(x)}}{4n_1^{(x)}n_2^{(x)} + n_1^{(x)} + n_2^{(x)}} \right\}^{\frac{1}{2}} \rho^{(x)},$$

$$\eta_3^{(x)} = \left[4 + \frac{1}{n_1^{(x)}} \frac{1}{n_2^{(x)}} \right]^{-\frac{1}{2}} \left[\frac{2(\mu_1^{(x)} + a\sigma^{(x)}) - (\lambda_{1r}^{(x)} + \lambda_{2s}^{(x)})}{\sigma^{(x)}} \right].$$

Hence

$$(4.6) \quad E[e_{12}] = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \sum_{r=0}^{n_1^{(x)}} \sum_{s=0}^{n_2^{(x)}} \binom{n_1^{(x)}}{r} \binom{n_2^{(x)}}{s} \\ \cdot p^{r+s} (1-p)^{n_1^{(x)} + n_2^{(x)} - r - s} [pH_1^*(\eta_1^{(x)}, \beta_1^{(x)}; \eta_2^{(x)}) \\ + (1-p)H_2^*(\eta_3^{(x)}, \beta_1^{(x)}; \eta_2^{(x)})]$$

where

$$H_1^* = H[\eta_1^{(x)}, \beta_1^{(x)}; \eta_2^{(x)}] + H[\eta_1^{(x)}, -\beta_1^{(x)}; -\eta_2^{(x)}]$$

$$H_2^* = H[\eta_3^{(x)}, \beta_1^{(x)}; \eta_2^{(x)}] + H[\eta_3^{(x)}, -\beta_1^{(x)}; -\eta_2^{(x)}]$$

Similarly,

$$(4.7) \quad E[e_{21}] = \sum_{x=0}^1 \theta^x (1-\theta)^{1-x} \sum_{r=0}^{n_1^{(x)}} \sum_{s=0}^{n_2^{(x)}} \binom{n_1^{(x)}}{r} \binom{n_2^{(x)}}{s} \\ \cdot p^{r+s} (1-p)^{n_1^{(x)}+n_2^{(x)}-r-s} [p H_3^*(\eta_1^{*(x)}, \beta_1^{(x)}; \eta_2^{*(x)}) \\ + (1-p) H_4^*(\eta_3^{*(x)}, \beta_1^{(x)}; \eta_2^{*(x)})]$$

where H_3^* , H_4^* are similar to H_1^* , H_2^* respectively; also the η^* 's are similar to η 's with $\mu_1^{(x)}$ being replaced by $\mu_2^{(x)}$

4.3 Numerical results

The basic model as well as the parameter structures are as given in section 3.3. Only two out of the four models are examined in detail. These are Model I and Model III. In addition for the nonnormality, three combinations of the mixture distributions are considered: (i) $p = 0.99$, $a = 0.25$, (ii) $p = 0.70$, $a = 0.25$, (iii) $p = 0.70$, $a = 1.00$. Case (i) is close to normality; case (iii) is one of extreme nonnormality. We have studied the effects of the nonnormality on the distribution function of e_{12} and on $E[e_{12}]$ and $E[e_{21}]$.

- (i) Distribution function. Figure 5 gives a comparison of $G_1(z)$ for the normal and the extreme nonnormal situations under Models I and III with the parameter $\theta = 0.5$ and the sample sizes $n_1 = n_2 = 20$.

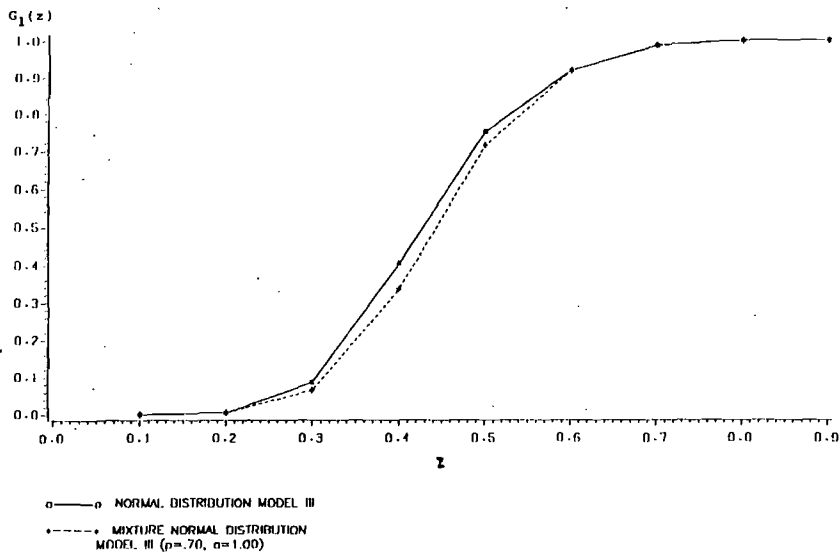
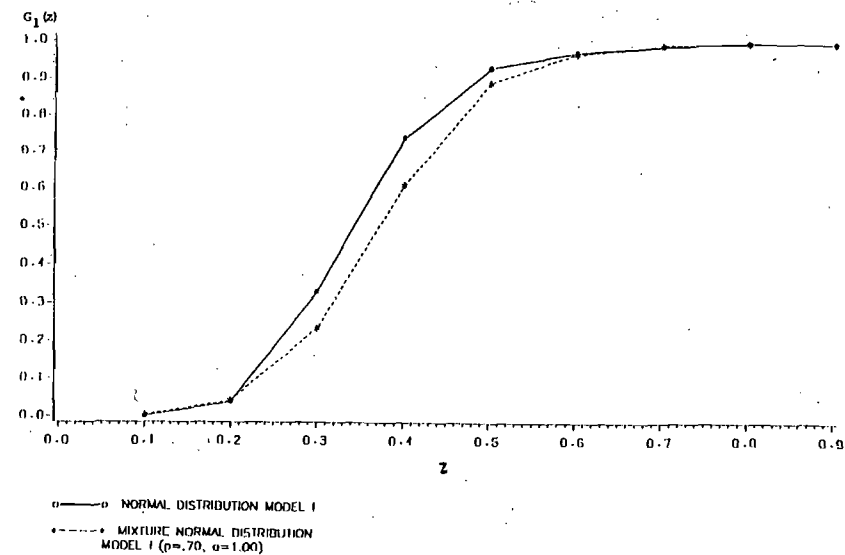


Figure 5. Distribution function $G_1(z)$ for $\theta = 0.50$ and $n_1 = n_2 = 20$

Table II. Expected values of the error rates for $n_1 = n_2 = 20$

θ	$E(e_{12})$ for Model I					$E(e_{12})$ for Model III					
	Normal	$p = 0,99$		$p = 0,70$		Normal	$p = 0,99$		$p = 0,70$		
		a		a			a		a		
		0,25	1,00	0,25	1,00		0,25	1,00	0,25	1,00	
0,25	0,333	0,335	0,336	0,336	0,349	0,432	0,429	0,429	0,429	0,436	
0,50	0,349	0,349	0,349	0,350	0,363	0,435	0,435	0,435	0,436	0,442	
0,75	0,365	0,363	0,363	0,364	0,376	0,438	0,436	0,436	0,436	0,443	
	$E(e_{21})$ for Model I					$E(e_{21})$ for Model III					
	0,25	0,333	0,335	0,336	0,336	0,355	0,432	0,429	0,429	0,429	0,442
	0,50	0,349	0,349	0,350	0,350	0,369	0,435	0,435	0,436	0,436	0,448
0,75	0,365	0,363	0,364	0,364	0,383	0,438	0,436	0,436	0,436	0,448	

The distribution of e_{12} in the mixtures is shifted to the right, compared with that of the normal case. This indicates that e_{12} tends to be larger (stochastically) in the mixtures. On the other hand the distribution of e_{12} (for the mixtures) is seen to be closer to that of the normal error rate, under Model III than under Model I. This is caused, in part, by the increased value of σ in Model III, dampening the effects of nonnormality. Also the error rates are higher for Model III (both in normal and nonnormal situations) than for Model I. This can be accounted for by the increase in σ^2 , resulting in a larger degree of overlap between the populations.

- (ii) Expected values. The expected values of e_{12} and e_{21} parallel each other. Although a better comparison of the behaviour of the error rates is through their distribution function, a summary of the information may be provided by their expected values. For Model I when $\theta < 0,5$, it appears that $E(e_{12})$ and $E(e_{21})$ are both smaller under normality than under nonnormality. For $\theta > 0,75$, this relationship holds only for severe nonnormality. However, for Model III any such comment can be made only for the extreme nonnormal situation. It should be pointed out that there are too many parameters affecting the behaviour of these expected values.

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